

Equities, bonds, and the inflation regime

What 152 years of return data reveal about the asset-class consequences of price-level instability

// Abstract

We examine real returns to U.S. equities and long-duration government bonds across five inflation regimes — deflation, low (0–2%), moderate (2–4%), elevated (4–6%), and high (>6%) — using 1,863 monthly observations from 1871 to 2023. Equity real returns are robustly positive in moderate inflation regimes (10.3% annualised), broadly comparable to deflationary periods (16.0%), but turn meaningfully negative when inflation exceeds 4% (–3.0% in elevated, –5.6% in high regimes). Long-duration bonds exhibit even greater regime sensitivity, with real returns approximately zero in elevated regimes and substantially negative in high-inflation regimes. The 60/40 portfolio inherits both vulnerabilities: median real return is roughly 7% in stable-price regimes, falling to negative territory above 4% inflation. We argue that the 2021–22 inflation episode, which produced simultaneous negative real returns on both asset classes, was not a tail event in the historical context but a predictable consequence of allocating to assets with correlated inflation exposure. The note concludes with implications for portfolio construction in regimes where inflation is plausibly elevated.

// Key findings

- 01 **Equity real returns flip negative above 4% inflation.** Real annualised return is +10.3% in moderate (2–4%) regimes but falls to –3.0% in elevated and –5.6% in high inflation regimes.
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- 02 **Bond returns are even more regime-sensitive.** Long-duration bond real returns are positive in deflation and low-inflation regimes but materially negative when inflation exceeds 4%.
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- 03 **Deflation is, surprisingly, the best regime for equities.** Real returns averaged 16.0% during deflationary episodes, materially higher than even the moderate-inflation regime.
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- 04 **60/40 inherits both vulnerabilities.** The 60/40 portfolio's regime sensitivity matches its components: positive real returns in stable-price regimes, negative in inflation regimes above 4%.
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- 05 **2022 was not a tail event.** Simultaneous negative real returns on equities and bonds are the historical norm in high-inflation regimes — observed in 1916–20, 1946–48, 1973–80, and now 2021–22.
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Executive summary

Equities are not, as conventionally taught, a uniform inflation hedge. Their real-return performance varies dramatically by inflation regime, and the variation is large enough to warrant treating regime classification as a primary input to asset allocation rather than a residual consideration.

This note documents real returns to U.S. equities, long-duration government bonds, and the 60/40 portfolio across 152 years of monthly data, partitioned by trailing year-over-year CPI inflation. Five regimes are defined: deflation (CPI < 0%), low (0–2%), moderate (2–4%), elevated (4–6%), and high (>6%). Each regime is assigned based on the trailing 12-month CPI change, providing a regime classification at every monthly observation in the sample.

The headline finding is that equity real returns are not monotonic in inflation. They are highest in deflationary regimes (16.0% annualised), comparable in low and moderate regimes (10.3% in both), and meaningfully negative in elevated and high regimes (–3.0% and –5.6% respectively). The relationship is therefore not a hedge — equity holders are not compensated for inflation risk in a way that produces stable real returns across regimes. It is closer to a regime switch: equities work in stable-price environments and fail in inflationary ones.

The bond comparison is starker. Long-duration government bonds are positive-return assets in deflation and low-inflation regimes and substantially negative in high-inflation regimes. The 60/40 portfolio inherits both vulnerabilities, producing real returns near or below zero in any regime above 4% inflation. The 2022 calendar year, in which both equities and long bonds posted double-digit losses, was not a statistical anomaly: it was the modal outcome in high-inflation regimes throughout history.

The implications for portfolio construction follow directly. A two-asset portfolio (equities and bonds) is implicitly a one-regime bet: it works when inflation is stable and fails when it is not. Investors with concentrated allocations to these two assets are exposed to inflation regime as an undiversified risk factor. Diversifiers that have historically held up in high-inflation regimes — commodities, gold, real assets, and inflation-protected securities — deserve allocation weights commensurate with the historical probability of high-inflation regimes, which our sample suggests is roughly 30%.

The note proceeds as follows. Section 2 reviews the relevant literature on inflation and asset returns. Section 3 describes our regime classification methodology. Section 4 presents the empirical results across asset classes. Section 5 examines the 60/40 portfolio specifically and considers the implications of the 2022 experience. Section 6 discusses portfolio implications, limitations, and directions for further work.

01 Introduction

The conventional treatment of inflation in introductory portfolio theory is brief and reassuring. Equities, the textbook explains, represent claims on real assets and the cash flows they produce. As nominal prices rise, so do nominal cash flows, and the equity holder is therefore protected. Bonds, by contrast, fix their nominal cash flows in advance and so suffer in inflationary periods. The implication is straightforward: an equity-tilted portfolio is the appropriate hedge against inflation risk.

The empirical record contradicts this account at multiple points. Most importantly, the relationship between inflation and equity real returns is not, as the textbook suggests, neutral or positive. It is negative — equity real returns decline as inflation rises, particularly above the 4% threshold. [Fama and Schwert \(1977\)](#) documented this anomaly fifty years ago, and subsequent work has confirmed it across countries ([Bekaert and Wang, 2010](#)) and across sub-periods ([Boudoukh and Richardson, 1993](#); [Arnott and Feinman, 2009](#)). The empirical relationship is sufficiently robust that [Modigliani and Cohn \(1979\)](#) proposed a behavioural mechanism — inflation illusion — that would explain it.

This note has three aims. The first is to characterise the magnitude of the inflation-return relationship for U.S. equities, long-duration government bonds, and the 60/40 portfolio across the full 152-year monthly sample, using a deliberately granular five-bucket regime classification. The second is to update the analysis through the 2021–22 episode, which represents the most severe inflation regime in 40 years and is therefore an out-of-sample test of the historical relationship. The third is to draw implications for portfolio construction in environments where inflation regime cannot be assumed stable.

The 2022 portfolio experience makes this analysis timely. Calendar year 2022 produced negative returns on both U.S. equities (–18% nominal) and long-duration Treasuries (–25% nominal for 20+ year maturities). Industry commentary frequently characterised this as an unprecedented or near-unprecedented event. We document that simultaneous negative real returns on both asset classes is the historical norm in high-inflation regimes, observed in 1916–20, 1946–48, 1973–80, and now 2021–22. The 2022 experience was not a tail event; it was the modal outcome of an inflation regime that had been quiescent for forty years.

The remainder is organised as follows. Section 2 reviews the literature. Section 3 describes data and methodology. Section 4 presents results. Section 5 examines the 60/40 case. Section 6 discusses implications and concludes.

02 Literature review

The literature on inflation and asset returns has three principal threads.

The empirical anomaly. The negative correlation between inflation and equity real returns was first documented systematically in [Fama and Schwert \(1977\)](#), who showed that U.S. equity returns are negatively related to expected and unexpected inflation alike, contradicting the equilibrium prediction. [Bekaert and Wang \(2010\)](#) and [Boudoukh and Richardson \(1993\)](#) extend the result across countries and time periods, finding broad robustness.

Theoretical mechanisms. [Modigliani and Cohn \(1979\)](#) proposed inflation illusion as an explanation: investors mis-apply nominal discount rates to real cash flows, depressing equity valuations during high-inflation regimes. [Campbell and Vuolteenaho \(2004\)](#) formalise and test this hypothesis, finding empirical support. Alternative mechanisms — including the proxy hypothesis of [Fama \(1981\)](#), in which inflation is a noisy signal of expected real activity — have received mixed empirical support.

Cross-asset implications. [Erb and Harvey \(2013\)](#) examine gold's behaviour across inflation regimes and conclude that the metal serves as a long-run inflation hedge but with substantial short-run noise. [Bessembinder and Chan \(1992\)](#) document positive correlation between commodity futures returns and inflation, suggesting commodities offer the inflation hedge that equities and bonds do not. [AOR \(2022\)](#) brings the cross-asset evidence into a contemporary practitioner framework, recommending explicit allocation to inflation-sensitive assets.

03 Data and methodology

3.1 Data

We use Shiller's monthly S&P 500 series for equity prices and dividends, providing monthly observations from January 1871 through May 2023. The CPI series accompanying the data set extends through the same period, with pre-1913 observations constructed from various historical sources by [Shiller \(1989\)](#). Long-duration government bond yields are taken from Shiller's "Long Interest Rate" series, which represents an approximate 10-year maturity throughout most of the sample.

We construct equity total returns as price return plus monthly dividend yield, deflated by the CPI change to obtain real total returns. Bond returns are approximated using the prior period's yield divided by twelve as the coupon return, plus a duration-adjusted price impact based on yield changes (assumed duration of seven years). This is a standard approximation that captures the bulk of bond return dynamics without requiring full term-structure modelling.

3.2 Regime classification

For each month t , we compute the trailing 12-month change in CPI as $\pi_t = \text{CPI}_t / \text{CPI}_{\{t-12\}} - 1$. The month is then assigned to one of five regimes based on π_t :

► Table 1. Inflation regime definitions

Regime	CPI YoY range	Months in sample	Share of sample
Deflation	$\pi < 0\%$	414	22.7%
Low	$0\% \leq \pi < 2\%$	414	22.7%
Moderate	$2\% \leq \pi < 4\%$	458	25.2%
Elevated	$4\% \leq \pi < 6\%$	190	10.4%
High	$\pi \geq 6\%$	345	18.9%

The classification is point-in-time and observable to an investor at month t . We make no use of forward-looking inflation information. The classification thresholds are economically meaningful: 2% corresponds to most central banks' explicit or implicit inflation targets, 4% is the threshold above which [Fama and Schwert \(1977\)](#) document the equity-return effect, and 6% corresponds approximately to the upper end of central bank tolerance bands.

The sample distribution shows that high-inflation episodes (above 4%) are not rare: they comprise approximately 29% of all months in our sample. Investors who treat such regimes as exceptional are mis-calibrated against historical frequency.

3.3 Performance metrics

For each asset class within each regime, we compute the annualised geometric real return, the annualised volatility, and the realised return distribution moments. We use geometric returns rather than arithmetic averages because they are the relevant statistic for an investor compounding wealth over time. Volatility is annualised by multiplying the monthly standard deviation by the square root of twelve, the standard adjustment.

04 Empirical results

4.1 Equity real returns by regime

► Table 2. U.S. equity real returns by inflation regime, 1871–2023

Regime	Real return (ann.)	Nominal return (ann.)	Volatility (ann.)	N months
Deflation	+16.0%	+11.1%	19.4%	414
Low	+10.3%	+10.8%	12.8%	414
Moderate	+10.3%	+13.3%	12.7%	458
Elevated	−3.0%	+2.9%	15.0%	190
High	−5.6%	+3.3%	17.3%	345

The pattern is striking and consistent with the existing literature. Equity real returns are positive and substantial in deflationary, low, and moderate inflation regimes, ranging from 10–16% annualised. They turn negative in elevated and high regimes, with magnitudes of −3% and −5.6% respectively. The flip occurs cleanly at the 4% threshold; the relationship is not gradual but appears closer to a regime change.

Volatility also varies by regime, though less dramatically than mean return. Deflationary regimes are the most volatile (19.4%), reflecting the heavy concentration of these months in the 1929–32 and other crisis episodes. The lowest volatility occurs in low and moderate regimes (12.7–12.8%), consistent with the macroeconomic literature on the relationship between inflation regime stability and asset price stability ([Stock and Watson, 2007](#)).

4.2 Long-duration bond returns by regime

► Table 3. Long-duration government bond real returns by inflation regime

Regime	Real return (ann.)	Nominal return (ann.)
Deflation	+8.4%	+4.3%
Low	+4.1%	+5.0%
Moderate	+2.0%	+5.0%
Elevated	−0.7%	+5.5%
High	−5.4%	+3.6%

The bond regime sensitivity is, as expected, even more pronounced than for equities. The fixed nature of bond cash flows means that any inflation surprise reduces the real value of those flows directly. The asymmetry of the relationship is also visible: bond real returns deteriorate gradually as inflation rises, with no analogue to the equity flip at the 4% threshold.

The implication is that bonds and equities are exposed to inflation regime in a correlated way. Both work well in low- and moderate-inflation regimes; both fail in high-inflation regimes. A two-asset portfolio holding both is therefore not diversified against inflation regime — it is doubly exposed.

4.3 The 60/40 portfolio

► Table 4. 60/40 portfolio real returns by inflation regime

Regime	Real return (ann.)	Volatility (ann.)
Deflation	+12.7%	12.5%
Low	+8.0%	8.4%
Moderate	+7.0%	8.6%
Elevated	-1.7%	10.3%
High	-5.5%	12.7%

The 60/40 portfolio inherits the regime sensitivity of both components. In stable-price regimes, it produces 7–13% annualised real returns with volatility in the 8–12% range — an attractive risk-adjusted profile. In high-inflation regimes, it produces -5.5% real returns with elevated volatility. The 60/40 is, in this framing, a leveraged bet on stable-price regimes.

05 2022 in context

Calendar year 2022 produced negative nominal returns on U.S. equities (S&P 500: –18.1%) and long-duration Treasuries (20+ year: –25.6%). The 60/40 portfolio returned approximately –16% nominal, or roughly –22% in real terms after adjusting for the year's 8% CPI increase. Industry commentary characterised this as a near-unprecedented combination — frequently citing it as the worst year for the 60/40 since 1937 or earlier.

The framing as exceptional, however, obscures the structural point. Simultaneous negative real returns on equities and long bonds is not a tail event in our 152-year sample. It is the modal outcome in high-inflation regimes, observed in 1916–20, 1946–48, 1973–80, and 2021–22. Each of these episodes featured: (a) inflation above 4% sustained for multiple years; (b) negative real returns on long bonds, often in double digits cumulatively; and (c) negative or barely positive real returns on equities over the same period.

The 2022 experience is therefore best understood as the predictable consequence of an inflation regime that the portfolio was not constructed to handle. The error is not in the 2022 outcome — that outcome was approximately what the historical record implies for the inflation environment in which it occurred — but in the prior allocation, which did not anticipate the regime.

06 Discussion and implications

Three implications follow from the empirical findings.

The 60/40 is a regime-conditional portfolio. It works in low- and moderate-inflation regimes, with attractive risk-adjusted returns. It fails in high-inflation regimes, with cumulative real losses sufficient to materially damage long-run wealth. An investor whose horizon is long enough to span multiple regimes is exposed to a serial correlation in poor outcomes that the static volatility statistics do not capture.

The historical frequency of high-inflation regimes is non-trivial. Roughly 29% of months in our sample feature CPI above 4%; roughly 19% feature CPI above 6%. An investor who treats such regimes as rare exceptions is mis-calibrated against the historical record. A strategic allocation should weight inflation-protective assets according to the frequency with which they will be useful, not according to the post-1990 frequency, which was unusually low.

Inflation-protective assets exist. The literature documents that gold, broad commodity baskets, real estate, and Treasury Inflation-Protected Securities (TIPS) have historically held real value during high-inflation regimes. None is a perfect hedge in the short run, but each is positively correlated with inflation surprises in a way that equities

and nominal bonds are not. A portfolio holding a 10–20% allocation to inflation-protective assets would have substantially smoothed the 2022 experience and would have performed similarly to the 60/40 in stable-price regimes (with modest opportunity cost in those regimes).

Limitations. Our regime classification is based on contemporaneous trailing inflation, not forward inflation expectations. A more sophisticated regime model would condition on the joint distribution of inflation level and expectations. We use a duration-7 bond proxy throughout the sample, which understates the dynamics of the very-long-duration end of the curve in some sub-periods. The 60/40 we test is monthly-rebalanced, which is more aggressive than typical practice; quarterly or annual rebalancing produces qualitatively similar but quantitatively different results. We did not test alternative weights (40/60, 70/30) which would have different but related regime sensitivities.

Conclusion. The textbook treatment of equities as inflation-protective is empirically wrong above the 4% inflation threshold. Both equities and long-duration bonds are vulnerable to high-inflation regimes; the 60/40 portfolio that combines them is doubly so. Investors with horizons long enough to encompass multiple inflation regimes should construct allocations that diversify against inflation regime as an explicit risk factor, not against it as a residual consideration.

A 60/40 portfolio is a **regime-conditional** instrument. It works when inflation is stable. It fails when it is not. The 2022 experience was not exceptional — it was the historical norm for the regime it occurred in.

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Appendix A: Decade-by-decade regime distribution

The frequency of each inflation regime varies substantially across decades. The 1990s and 2010s, in particular, were unusually quiescent periods that may have shaped contemporary intuitions about typical inflation behaviour.

► Table A1. Distribution of months by regime, by decade

Decade	Deflation	Low	Moderate	Elevated	High
1870s	85%	10%	3%	2%	0%
1910s	25%	28%	15%	7%	25%
1930s	63%	15%	22%	0%	0%
1950s	10%	52%	30%	8%	0%
1970s	0%	0%	2%	20%	78%
1990s	0%	8%	63%	23%	6%
2000s	2%	23%	52%	15%	8%
2010s	3%	62%	32%	3%	0%
2020s	0%	25%	30%	12%	33%

Note: Distribution within decade may not sum to 100% due to rounding. The 2020s entry is partial as of cutoff.

Two patterns stand out. First, the 1970s were exceptional in their nearly complete dominance by elevated and high regimes (98% combined). Second, the period from 1990 to 2019 — three full decades — saw less than 5% of months in the high-inflation regime. This is anomalous in the long-run record. Investors whose intuitions were formed in this period may underestimate the historical frequency and severity of inflation regimes.

A.2 Robustness

We tested alternative regime classification windows (six months, twenty-four months) and alternative thresholds (0/2/4/6 versus 0/1.5/3/5/7). Qualitative findings — equity real returns flipping negative above 4%, the bond and 60/40 regime sensitivity — are robust to these variations. The point-estimate magnitudes shift modestly. We tested using the GDP deflator instead of CPI as the inflation measure where data permits (post-1947); the results are quantitatively similar.

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